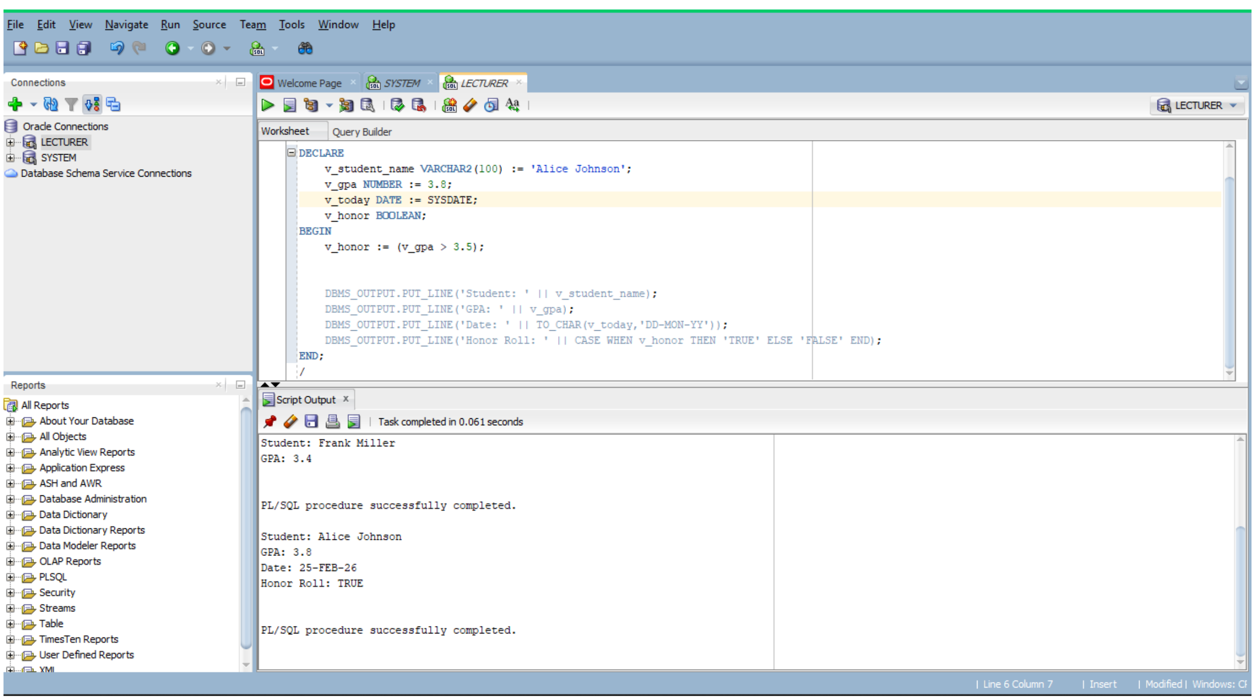
PART1

Q1.1

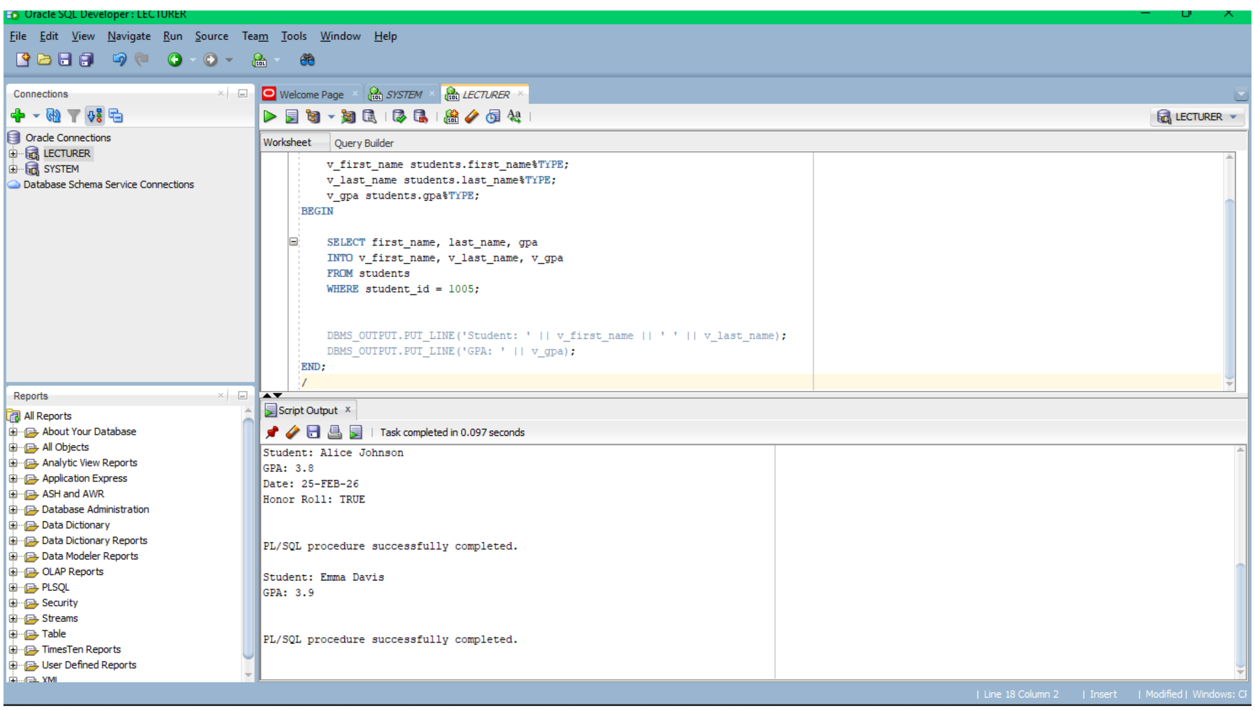
```
SET SERVEROUTPUT ON;

DECLARE
    v_student_name VARCHAR2(100) := 'Alice Johnson';
    v_gpa NUMBER := 3.8;
    v_today DATE := SYSDATE;
    v_honor BOOLEAN;
BEGIN
    v_honor := (v_gpa > 3.5);


    DBMS_OUTPUT.PUT_LINE('Student: ' || v_student_name);
    DBMS_OUTPUT.PUT_LINE('GPA: ' || v_gpa);
    DBMS_OUTPUT.PUT_LINE('Date: ' || TO_CHAR(v_today,'DD-MON-YY'));
    DBMS_OUTPUT.PUT_LINE('Honor Roll: ' || CASE WHEN v_honor THEN 'TRUE' ELSE
'FALSE' END);
END;
/
```



Q1.2

```
SET SERVEROUTPUT ON;
```

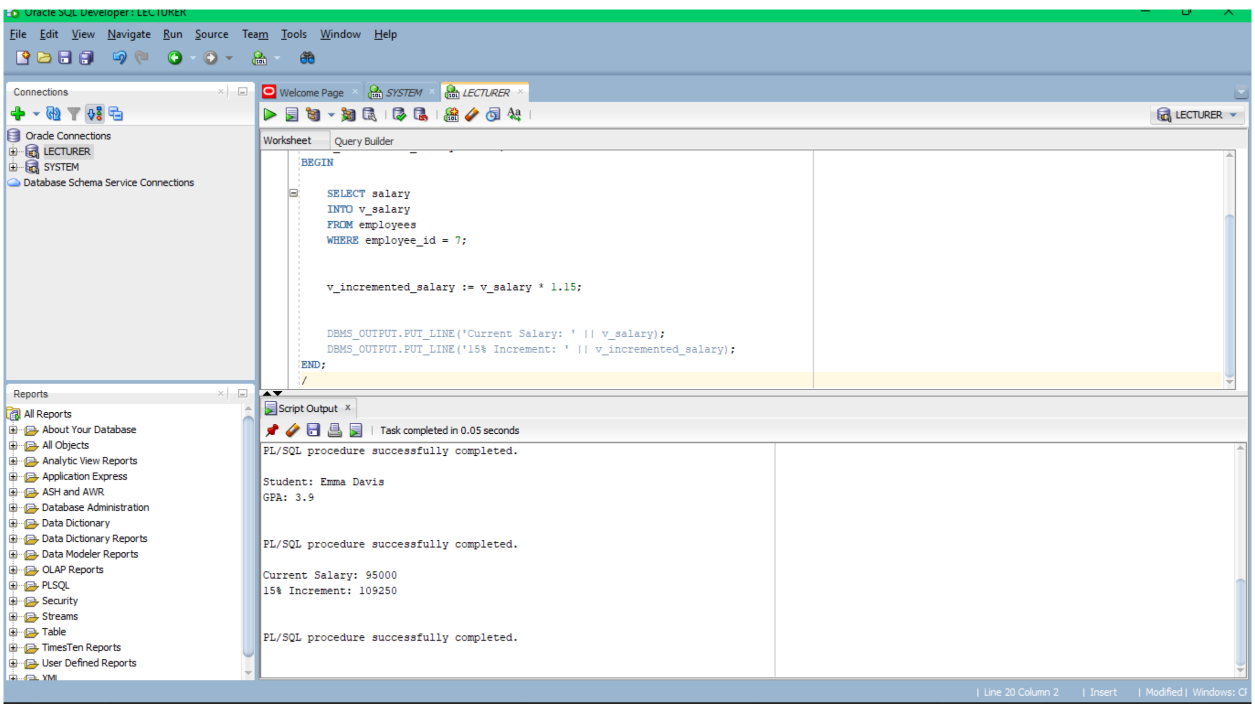
```
DECLARE
    v_first_name students.first_name%TYPE;
    v_last_name students.last_name%TYPE;
    v_gpa students.gpa%TYPE;
BEGIN

    SELECT first_name, last_name, gpa
    INTO v_first_name, v_last_name, v_gpa
    FROM students
    WHERE student_id = 1005;


    DBMS_OUTPUT.PUT_LINE('Student: ' || v_first_name || ' ' || v_last_name);
    DBMS_OUTPUT.PUT_LINE('GPA: ' || v_gpa);
END;
/
```



Q1.3

```
SET SERVEROUTPUT ON;

DECLARE
    v_salary employees.salary%TYPE;
    v_incremented_salary NUMBER;
BEGIN
```

```
    SELECT salary
    INTO v_salary
    FROM employees
    WHERE employee_id = 7;


    v_incremented_salary := v_salary * 1.15;


    DBMS_OUTPUT.PUT_LINE('Current Salary: ' || v_salary);
    DBMS_OUTPUT.PUT_LINE('15% Increment: ' || v_incremented_salary);
END;
/
```

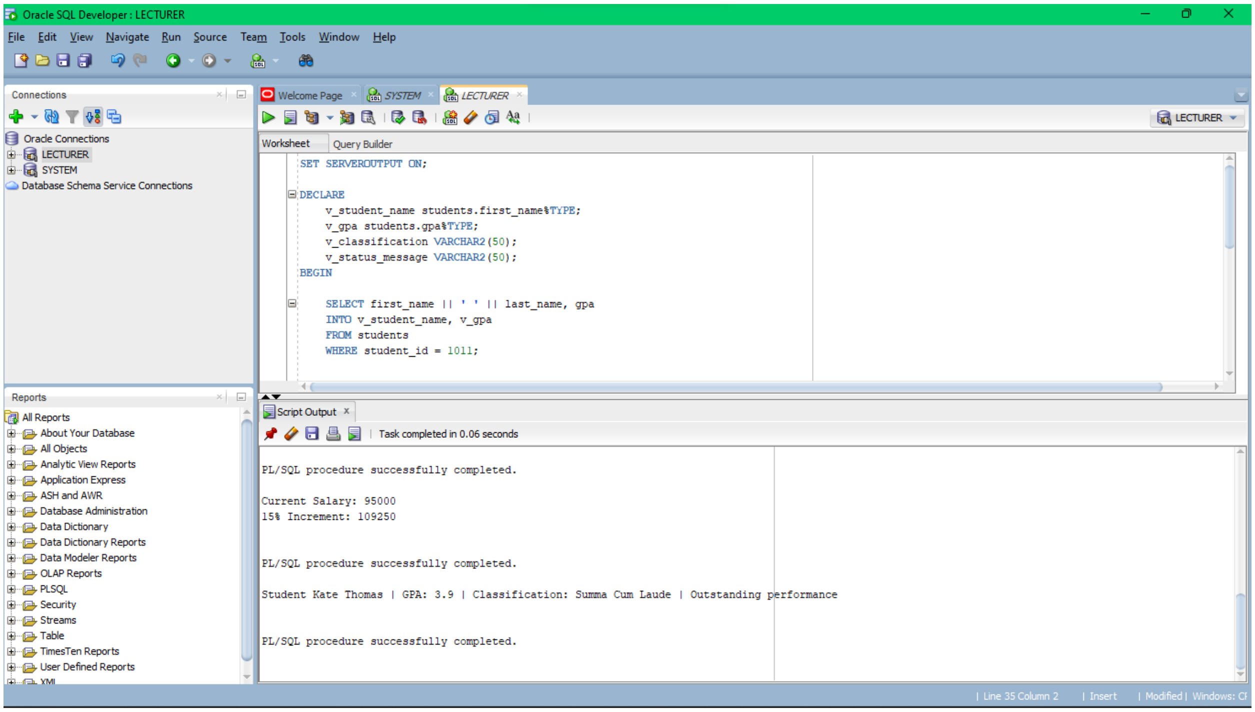


PART2

Q2.1

```
SET SERVEROUTPUT ON;

DECLARE
    v_student_name students.first_name%TYPE;
    v_gpa students.gpa%TYPE;
    v_classification VARCHAR2(50);
    v_status_message VARCHAR2(50);
```

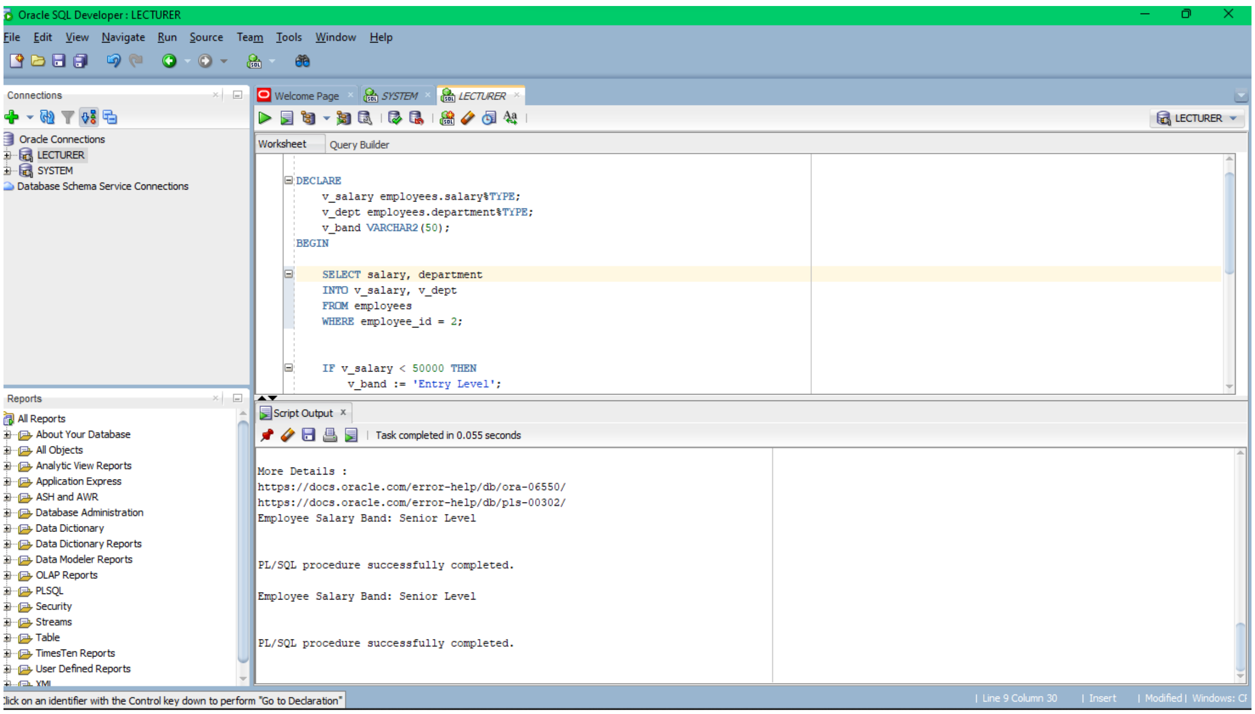
```sql
BEGIN

    SELECT first_name || ' ' || last_name, gpa
    INTO v_student_name, v_gpa
    FROM students
    WHERE student_id = 1011;


    IF v_gpa BETWEEN 3.7 AND 4 THEN
        v_classification := 'Summa Cum Laude';
        v_status_message := 'Outstanding performance';
    ELSIF v_gpa BETWEEN 3.3 AND 3.6 THEN
        v_classification := 'Magna Cum Laude';
        v_status_message := 'Outstanding performance';
    ELSIF v_gpa BETWEEN 3.0 AND 3.2 THEN
        v_classification := 'Cum Laude';
        v_status_message := 'Excellent performance';
    ELSIF v_gpa BETWEEN 2.0 AND 2.9 THEN
        v_classification := 'Satisfactory';
        v_status_message := 'Good performance';
    ELSE
        v_classification := 'Academic Probation';
        v_status_message := 'Urgent action required';
    END IF;

    DBMS_OUTPUT.PUT_LINE('Student ' || v_student_name || ' | GPA: ' || v_gpa || '
| Classification: ' || v_classification || ' | ' || v_status_message);
END;
/
```



Q2.2

```sql
SET SERVEROUTPUT ON;

DECLARE
    v_salary employees.salary%TYPE;
    v_dept employees.department%TYPE;
    v_band VARCHAR2(50);
BEGIN

    SELECT salary, department
    INTO v_salary, v_dept
    FROM employees
    WHERE employee_id = 2;


    IF v_salary < 50000 THEN
        v_band := 'Entry Level';
    ELSIF v_salary BETWEEN 50000 AND 69999 THEN
        v_band := 'Mid Level';
    ELSIF v_salary BETWEEN 70000 AND 89999 THEN
        v_band := 'Senior Level';
    ELSE
        v_band := 'Executive Level';
    END IF;
```

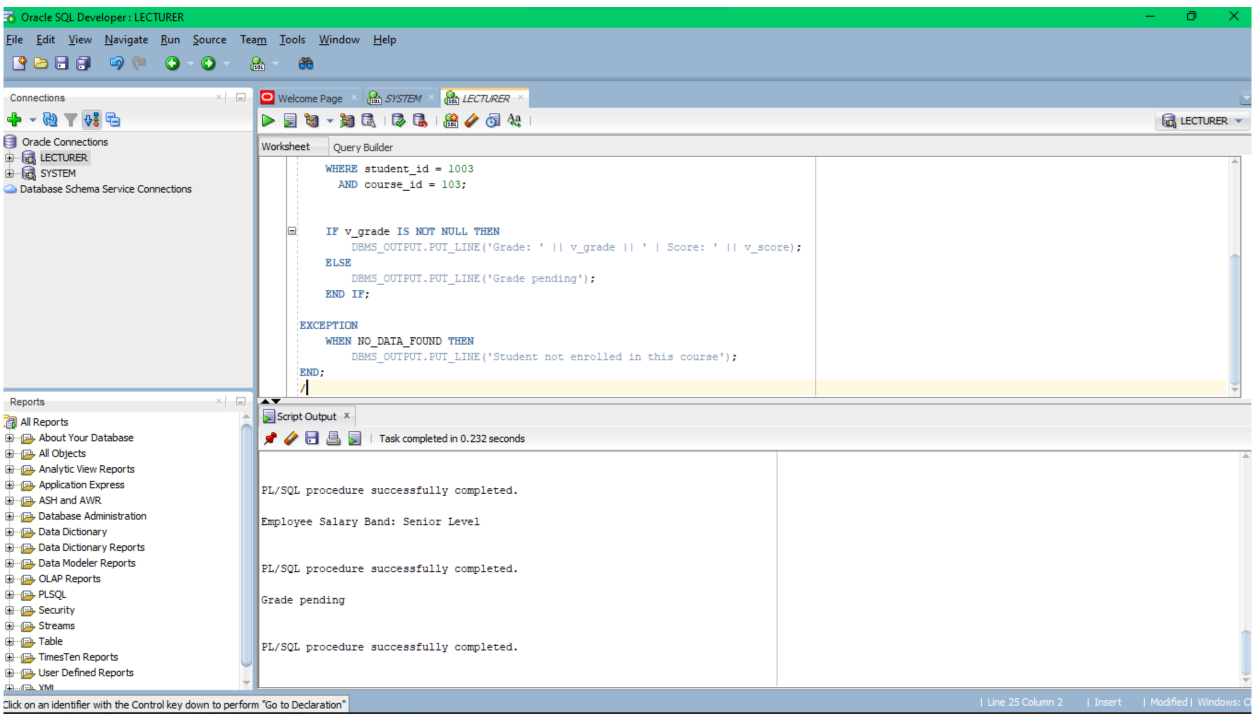
```
    DBMS_OUTPUT.PUT_LINE('Employee Salary Band: ' || v_band);


    IF v_dept = 'Computer Science' AND (v_band = 'Senior Level' OR v_band =
'Executive Level') THEN
        DBMS_OUTPUT.PUT_LINE('Eligible for Tech Leadership Program');
    END IF;
END;
/
```



Q2.3

```
SET SERVEROUTPUT ON;

DECLARE
    v_grade enrollments.grade%TYPE;
    v_score enrollments.score%TYPE;
BEGIN

    SELECT grade, score
    INTO v_grade, v_score
    FROM enrollments
    WHERE student_id = 1003
      AND course_id = 103;


```

```
    IF v_grade IS NOT NULL THEN
        DBMS_OUTPUT.PUT_LINE('Grade: ' || v_grade || ' | Score: ' || v_score);
    ELSE
        DBMS_OUTPUT.PUT_LINE('Grade pending');
    END IF;

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Student not enrolled in this course');
END;
/
```

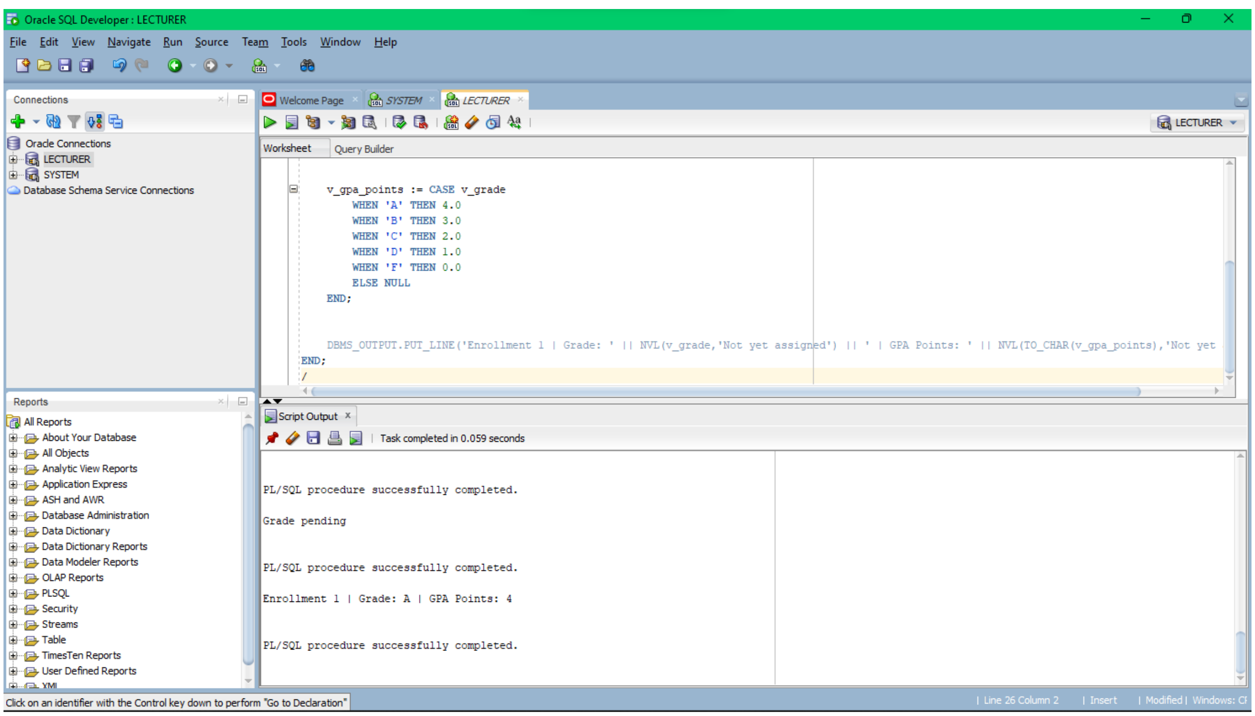


PART3

Q3.1

```
SET SERVEROUTPUT ON;

DECLARE
    v_grade enrollments.grade%TYPE;
    v_gpa_points NUMBER;
BEGIN

    SELECT grade
```

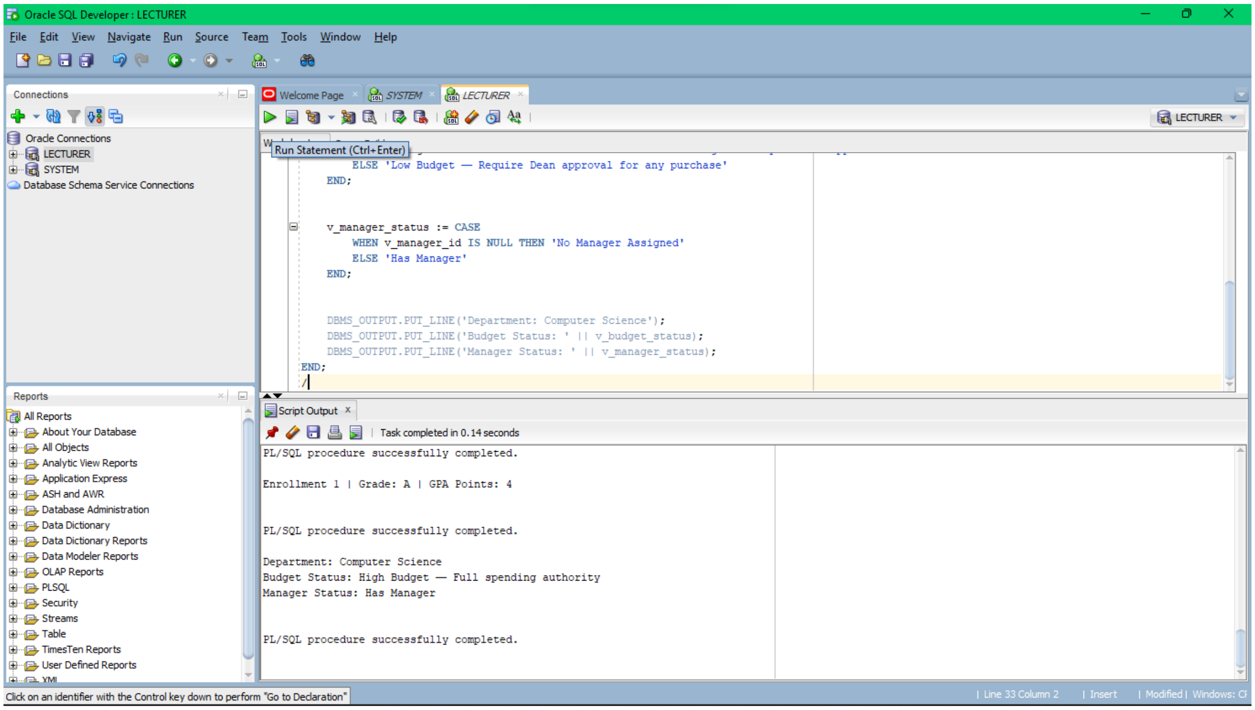
```
    INTO v_grade
    FROM enrollments
    WHERE enrollment_id = 1;


    v_gpa_points := CASE v_grade
        WHEN 'A' THEN 4.0
        WHEN 'B' THEN 3.0
        WHEN 'C' THEN 2.0
        WHEN 'D' THEN 1.0
        WHEN 'F' THEN 0.0
        ELSE NULL
    END;


    DBMS_OUTPUT.PUT_LINE('Enrollment 1 | Grade: ' || NVL(v_grade,'Not yet
assigned') || ' | GPA Points: ' || NVL(TO_CHAR(v_gpa_points),'Not yet
assigned'));
END;
/
```



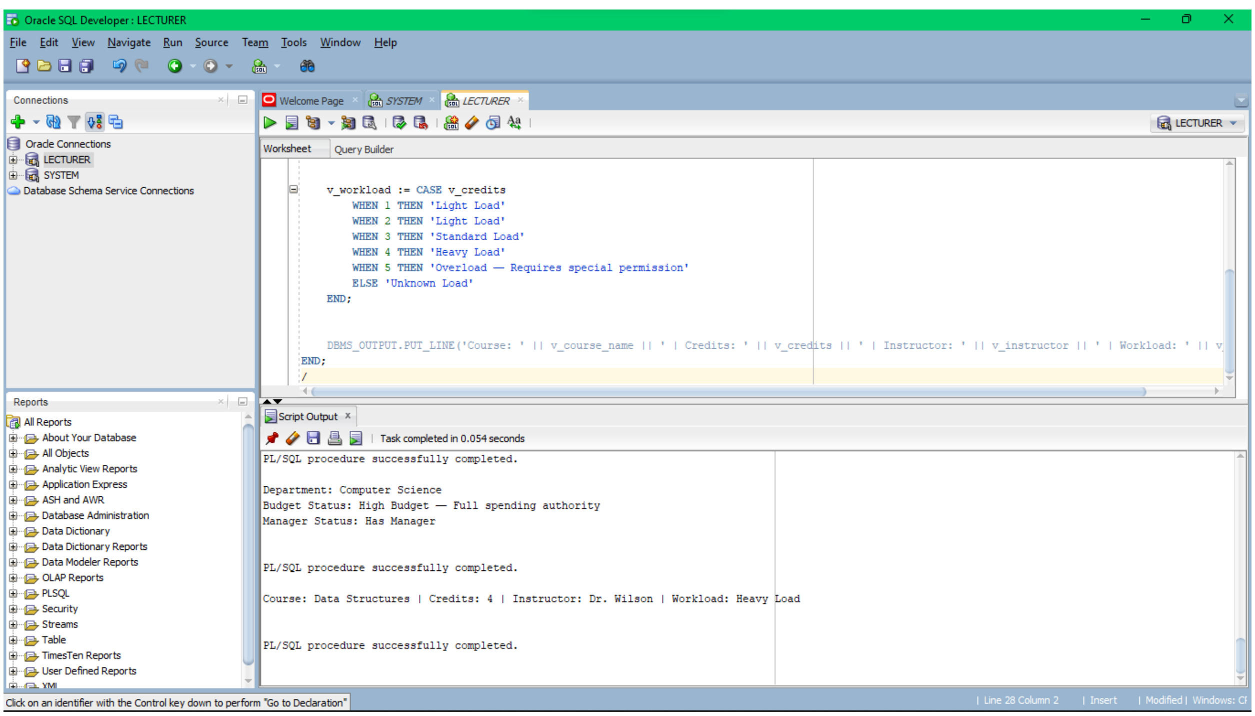
Q3.2

```
SET SERVEROUTPUT ON;

DECLARE
```

```
    v_budget departments.budget%TYPE;
    v_manager_id departments.manager_id%TYPE;
    v_budget_status VARCHAR2(100);
    v_manager_status VARCHAR2(50);
BEGIN

    SELECT budget, manager_id
    INTO v_budget, v_manager_id
    FROM departments
    WHERE department_name = 'Computer Science';


    v_budget_status := CASE
        WHEN v_budget > 700000 THEN 'High Budget – Full spending authority'
        WHEN v_budget BETWEEN 500000 AND 700000 THEN 'Medium Budget – Require HOD
approval above 50k'
        ELSE 'Low Budget – Require Dean approval for any purchase'
    END;


    v_manager_status := CASE
        WHEN v_manager_id IS NULL THEN 'No Manager Assigned'
        ELSE 'Has Manager'
    END;


    DBMS_OUTPUT.PUT_LINE('Department: Computer Science');
    DBMS_OUTPUT.PUT_LINE('Budget Status: ' || v_budget_status);
    DBMS_OUTPUT.PUT_LINE('Manager Status: ' || v_manager_status);
END;
/
```

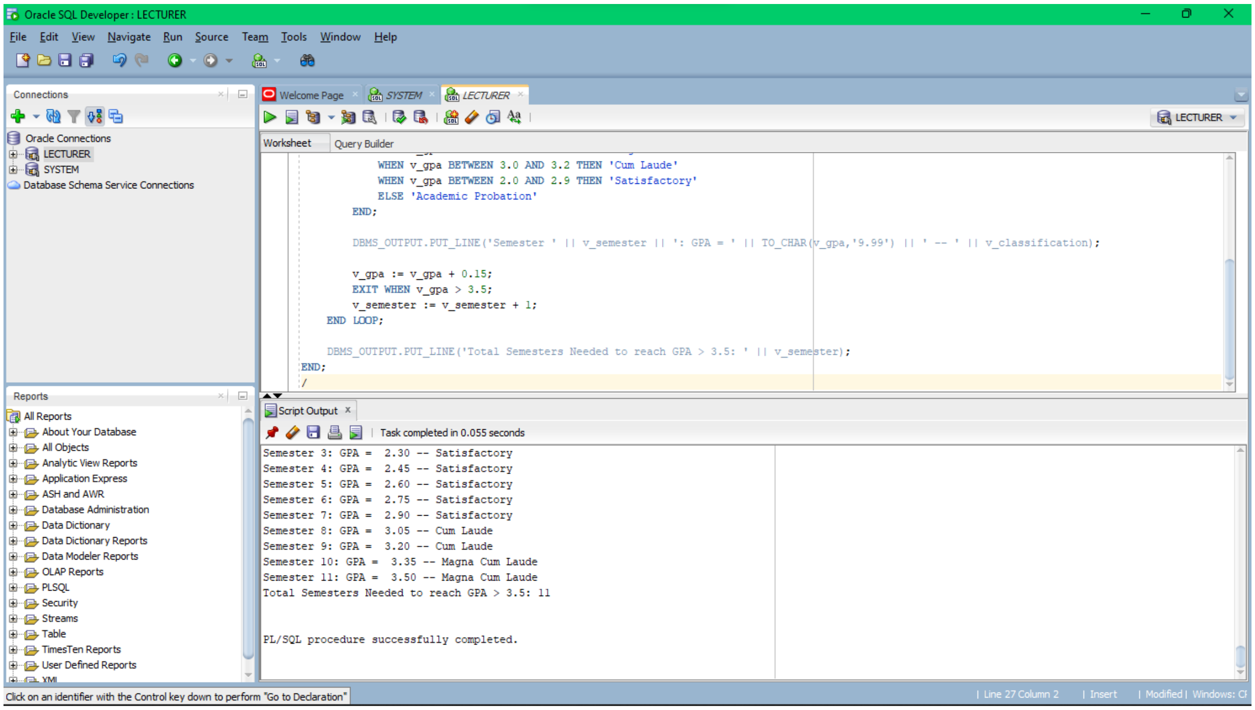


Q3.3



PART4

Q4.1



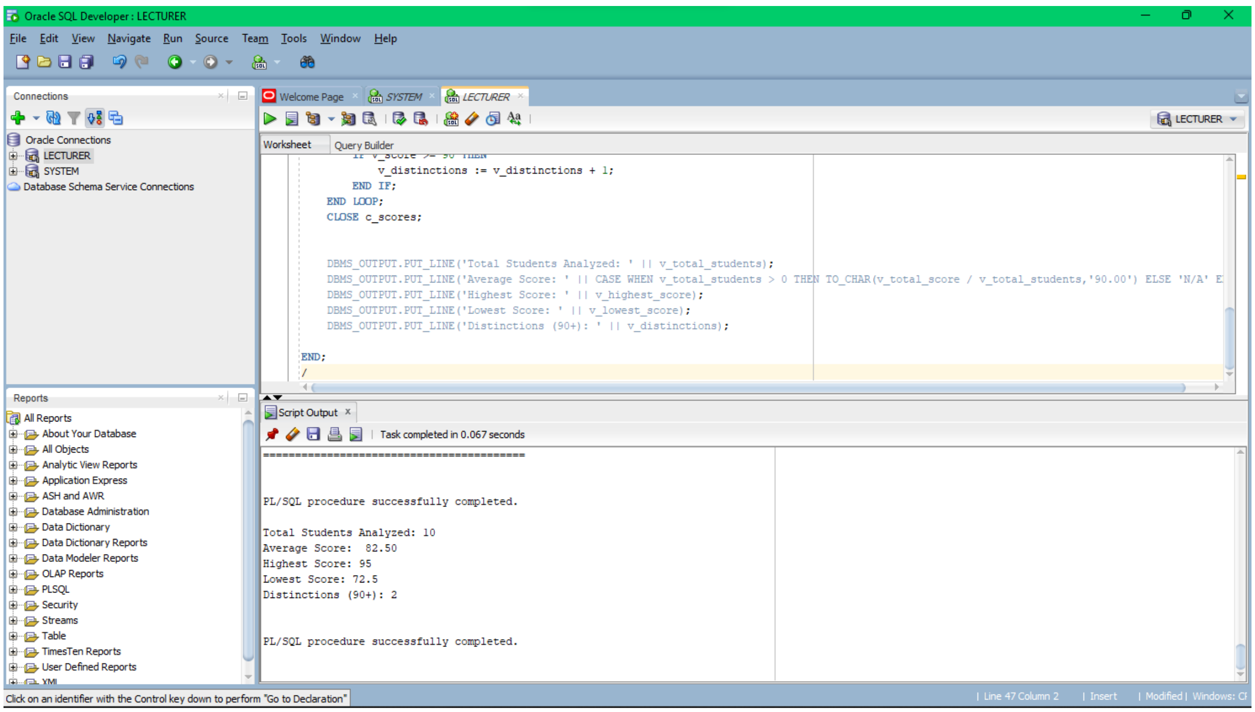

```
SET SERVEROUTPUT ON;

DECLARE
    v_gpa NUMBER := 2.0;
    v_semester NUMBER := 1;
    v_classification VARCHAR2(50);
BEGIN
    LOOP

        v_classification := CASE
            WHEN v_gpa BETWEEN 3.7 AND 4.0 THEN 'Summa Cum Laude'
            WHEN v_gpa BETWEEN 3.3 AND 3.6 THEN 'Magna Cum Laude'
            WHEN v_gpa BETWEEN 3.0 AND 3.2 THEN 'Cum Laude'
            WHEN v_gpa BETWEEN 2.0 AND 2.9 THEN 'Satisfactory'
            ELSE 'Academic Probation'
        END;

        DBMS_OUTPUT.PUT_LINE('Semester ' || v_semester || ': GPA = ' ||
TO_CHAR(v_gpa,'9.99') || ' -- ' || v_classification);

        v_gpa := v_gpa + 0.15;
        EXIT WHEN v_gpa > 3.5;
        v_semester := v_semester + 1;
    END LOOP;
```

```
    DBMS_OUTPUT.PUT_LINE('Total Semesters Needed to reach GPA > 3.5: ' ||
v_semester);
END;
/
```

Q4.2

```
SET SERVEROUTPUT ON;

DECLARE
    CURSOR c_scores IS
        SELECT score
        FROM enrollments
        WHERE course_id = 101
        AND score IS NOT NULL;

    v_score enrollments.score%TYPE;
    v_total_students NUMBER := 0;
    v_total_score NUMBER := 0;
    v_highest_score NUMBER := -1;
    v_lowest_score NUMBER := 101;
    v_distinctions NUMBER := 0;
BEGIN
    OPEN c_scores;
    LOOP
        FETCH c_scores INTO v_score;
        EXIT WHEN c_scores%NOTFOUND;

        v_total_students := v_total_students + 1;
        v_total_score := v_total_score + v_score;

        IF v_score > v_highest_score THEN
            v_highest_score := v_score;
        END IF;

        IF v_score < v_lowest_score THEN
            v_lowest_score := v_score;
        END IF;

        IF v_score >= 90 THEN
            v_distinctions := v_distinctions + 1;
        END IF;
    END LOOP;
    CLOSE c_scores;
```

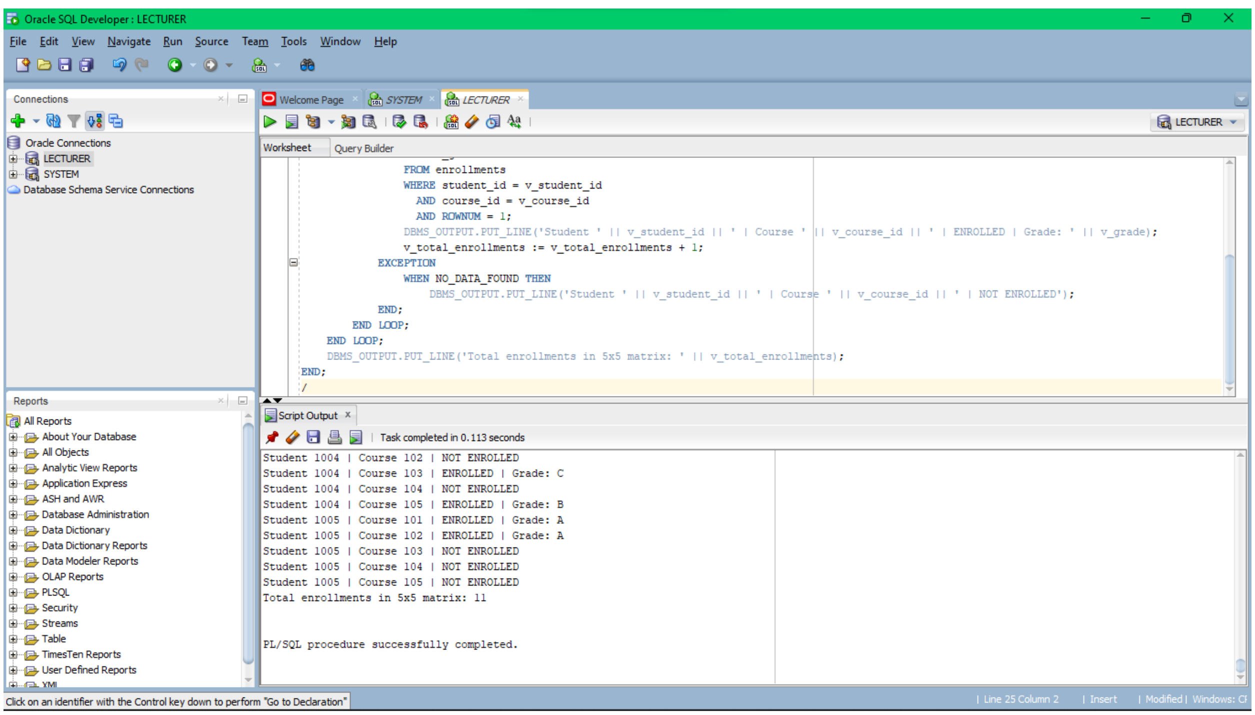
```
    DBMS_OUTPUT.PUT_LINE('Total Students Analyzed: ' || v_total_students);
    DBMS_OUTPUT.PUT_LINE('Average Score: ' || CASE WHEN v_total_students > 0 THEN TO_CHAR(v_total_score / v_total_students,'90.00') ELSE 'N/A' END);
    DBMS_OUTPUT.PUT_LINE('Highest Score: ' || v_highest_score);
    DBMS_OUTPUT.PUT_LINE('Lowest Score: ' || v_lowest_score);
    DBMS_OUTPUT.PUT_LINE('Distinctions (90+): ' || v_distinctions);

END;
/
```



Q4.3

```
SET SERVEROUTPUT ON;
DECLARE
    v_grade             VARCHAR2(2);
    v_total_enrollments NUMBER := 0;
BEGIN
    FOR v_student_id IN 1001..1005 LOOP
        FOR v_course_id IN 101..105 LOOP
            BEGIN
                SELECT NVL(grade, '-')
                INTO v_grade
                FROM enrollments
                WHERE student_id = v_student_id
```

```
                  AND course_id = v_course_id
                  AND ROWNUM = 1;
                DBMS_OUTPUT.PUT_LINE('Student ' || v_student_id || ' | Course ' || v_course_id || ' | ENROLLED | Grade: ' || v_grade);
                v_total_enrollments := v_total_enrollments + 1;
            EXCEPTION
                WHEN NO_DATA_FOUND THEN
                    DBMS_OUTPUT.PUT_LINE('Student ' || v_student_id || ' | Course ' || v_course_id || ' | NOT ENROLLED');
            END;
        END LOOP;
    END LOOP;
    DBMS_OUTPUT.PUT_LINE('Total enrollments in 5x5 matrix: ' || v_total_enrollments);
END;
/
```



PART5

THE SENARIO

```
SET SERVEROUTPUT ON;

DECLARE
```

```
    v_student_name students.first_name%TYPE;
    v_gpa students.gpa%TYPE;
    v_status students.status%TYPE;
    v_enroll_date students.enrollment_date%TYPE;
    v_years_enrolled NUMBER;
    v_continuing_status VARCHAR2(20);


    CURSOR c_courses IS
        SELECT e.course_id, c.course_name, c.credits, e.score, e.grade
        FROM enrollments e
        JOIN courses c ON e.course_id = c.course_id
        WHERE e.student_id = 1007;

    v_course c_courses%ROWTYPE;
    v_total_courses NUMBER := 0;
    v_total_credits NUMBER := 0;
    v_highest_score NUMBER := -1;
    v_best_course VARCHAR2(100) := '';
    v_in_progress NUMBER := 0;


    v_academic_standing VARCHAR2(50);
    v_scholarship_note VARCHAR2(100) := '';
BEGIN

    SELECT first_name || ' ' || last_name, gpa, status, enrollment_date
    INTO v_student_name, v_gpa, v_status, v_enroll_date
    FROM students
    WHERE student_id = 1007;


    v_years_enrolled := ROUND(MONTHS_BETWEEN(SYSDATE, v_enroll_date)/12,1);


    IF v_years_enrolled > 1 THEN
        v_continuing_status := 'Continuing Student';
    ELSE
        v_continuing_status := 'New Student';
    END IF;


    v_academic_standing := CASE
        WHEN v_gpa BETWEEN 3.7 AND 4 THEN 'Summa Cum Laude'
        WHEN v_gpa BETWEEN 3.3 AND 3.6 THEN 'Magna Cum Laude'
```

```
        WHEN v_gpa BETWEEN 3.0 AND 3.2 THEN 'Cum Laude'
        WHEN v_gpa BETWEEN 2.0 AND 2.9 THEN 'Satisfactory'
        ELSE 'Academic Probation'
    END;


    IF v_gpa >= 3.5 THEN
        v_scholarship_note := 'May qualify for merit scholarship';
    END IF;


    OPEN c_courses;
    LOOP
        FETCH c_courses INTO v_course;
        EXIT WHEN c_courses%NOTFOUND;

        v_total_courses := v_total_courses + 1;
        v_total_credits := v_total_credits + v_course.credits;


        IF v_course.score IS NOT NULL AND v_course.score > v_highest_score THEN
            v_highest_score := v_course.score;
            v_best_course := v_course.course_name;
        END IF;


        IF v_course.grade IS NULL THEN
            v_in_progress := v_in_progress + 1;
        END IF;
    END LOOP;
    CLOSE c_courses;



    DBMS_OUTPUT.PUT_LINE('SEMESTER PERFORMANCE REPORT');
    DBMS_OUTPUT.PUT_LINE('AUCA – DB Development with PL/SQL');
    DBMS_OUTPUT.PUT_LINE('Student: ' || v_student_name || ' (ID: 1007)');
    DBMS_OUTPUT.PUT_LINE('Status: ' || v_status || ' | ' || v_continuing_status);
    DBMS_OUTPUT.PUT_LINE('Current GPA: ' || TO_CHAR(v_gpa,'9.99') || ' | 
Standing: ' || v_academic_standing);
    IF v_scholarship_note IS NOT NULL THEN
        DBMS_OUTPUT.PUT_LINE('Scholarship Note: ' || v_scholarship_note);
    END IF;
```

```sql
    DBMS_OUTPUT.PUT_LINE('Courses Enrolled: ' || v_total_courses);
    DBMS_OUTPUT.PUT_LINE('Total Credits: ' || v_total_credits);
    DBMS_OUTPUT.PUT_LINE('Best Course: ' || NVL(v_best_course,'N/A') || ' (Score: ' || v_highest_score || ')');
    DBMS_OUTPUT.PUT_LINE('Courses In Progress: ' || v_in_progress);

END;
/
```